Capital and Storage Constrained Resilience: A Scenario-Based Agent-Based Simulation of Supplier Diversification and Inventory Buffering Strategies in the Philippine Food Manufacturing Context

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***Abstract*—** Small and medium enterprises (SMEs) dominate the Philippine food manufacturing sector, yet face binding capital and storage constraints that fundamentally alter which supply chain resilience strategy is feasible when disruptions occur. This study develops a scenario-based agent-based model (ABM) comparing supplier diversification against inventory buffering under matched conditions, calibrated to real Philippine Statistics Authority (PSA) Monthly Integrated Survey of Selected Industries (MISSI) production data and DTI/PSA MSME firm-size classifications. A twelve-parameter scenario space is sampled via Latin Hypercube Sampling, evaluated through Monte Carlo replication, and analyzed with a Random Forest model to identify the dominant drivers of strategy preference. A reduced-scale demonstration run of 100 scenarios at 20 Monte Carlo repetitions finds that disruption duration is the dominant driver of strategy preference (60.3 percent feature importance, R-squared 0.916): buffering is preferred in short and moderate disruptions, while diversification only wins scenarios where disruption duration exceeds approximately four months. Storage capacity shows a secondary, weaker effect in the same direction. These findings are preliminary, scoped to a reduced run with explicitly documented parameter-sourcing and tooling substitutions, and are intended to precede, not replace, the full 1,000-scenario, 100-repetition study specified in the original methodology.

***Index Terms*—**agent-based simulation, supply chain resilience, supplier diversification, inventory buffering, Monte Carlo simulation, Random Forest, SME food manufacturing, Philippines.

I. INTRODUCTION

A. Background and Context

Supply chain disruptions, whether from pandemics, natural disasters, or geopolitical shocks, force manufacturing firms to choose between two broad resilience strategies: diversifying their supplier base to reduce dependency on any single source, or holding larger inventory buffers to absorb temporary supply interruptions. Both strategies carry real costs. Diversification requires establishing and maintaining relationships with multiple suppliers, which is costly and administratively demanding. Inventory buffering requires capital tied up in stock and physical storage space to hold it.

For large manufacturers, these costs are manageable relative to available resources. For small and medium enterprises (SMEs), which dominate the Philippine food manufacturing sector, capital and storage capacity are binding constraints that fundamentally change which strategy is actually feasible, let alone optimal. A firm with limited capital cannot simply diversify to five suppliers; a firm with limited storage cannot simply stockpile six months of raw materials.

B. Problem Statement

This study asks: how do capital and storage capacity constraints affect the relative effectiveness of supplier diversification versus inventory buffering in reducing unmet demand and recovery time under disruptions of varying severity and duration, and does the preferred strategy shift as these constraints tighten?

Existing agent-based simulation literature on supply chain resilience has examined disruption recovery strategies, supplier diversification, and inventory policy, but not the specific combination this study requires. Four elements must be addressed together, and no reviewed study addresses all four: explicit capital and storage capacity constraints as the mechanism governing which strategy is even feasible; a systematic, controlled comparison of supplier diversification versus inventory buffering under matched conditions; varying disruption severity and duration rather than a single disruption profile; and situating the simulation in a verified Philippine industry context using real published data.

Compounding this gap is a data availability problem: no public source provides firm-level SME data on inventory, supplier behavior, or capital position for the Philippines. Any Philippine-situated simulation of this mechanism must therefore combine industry-level empirical context with literature-derived and officially classified, explicitly tested assumptions for firm-level mechanisms.

C. Chosen SDG

This research aligns primarily with SDG 9: Industry, Innovation, and Infrastructure, by helping food manufacturing SMEs make more informed resilience investment decisions under real resource constraints. It is also indirectly relevant to SDG 2, SDG 8, and SDG 12 through

supply chain continuity, SME job protection, and resource-efficient resilience planning respectively.

D. Research Questions

Main Research Question: How do capital and storage capacity constraints affect the relative effectiveness of supplier diversification versus inventory buffering in reducing unmet demand and recovery time under disruptions of varying severity and duration, and does the preferred strategy shift as these constraints tighten?

- RQ1: Under high, medium, and low capital constraint scenarios, how does relative performance change?
- RQ2: Under varying storage constraints, at what point does buffering's effectiveness decline?
- RQ3: How does disruption severity interact with resource constraints?
- RQ4: How does disruption duration interact with resource constraints?
- RQ5: Is there an identifiable crossover region, and how sensitive is it to cost and reliability assumptions?

E. Research Objectives

- Design and implement a scenario-based ABM of an SME food manufacturing firm under capital and storage constraints.
- Implement diversification and buffering as symmetrically-constrained behaviors applied to matched scenarios.
- Generate a synthetic dataset via Latin Hypercube Sampling and Monte Carlo replication.
- Identify, using Random Forest and SHAP analysis, the conditions under which each strategy wins.
- Independently audit and validate the model's mechanics and data integrity prior to drawing conclusions.

II. REVIEW OF RELATED LITERATURE

A. Agent-Based Approaches to Disruption and Resilience

Li and Coates [1] developed an ABM for business operations under flood disruption, establishing ABM viability for firm-level disruption response. Lohmer, Bugert, and Lasch [2] analyzed resilience strategies and ripple effects in blockchain-coordinated supply chains, demonstrating how strategy choice propagates through a network under disruption. Zhu, Han, and Liu [3] modeled adaptive cooperation among heterogeneous manufacturers under fluctuating post-COVID-19 demand, showing firm heterogeneity materially affects simulated outcomes.

B. Hybrid and Multi-Method Frameworks

Anagnostou, Mintram, and Taylor [4] proposed SCROAM, a hybrid ABM plus early-warning-system framework analogous to the ABM-plus-ML pipeline adopted here. Bi, Chen, Tilbury, Shen, and Barton [5] presented a model-based multi-agent framework for agile disruption response, contributing agent decision-rule design patterns.

C. ABM Combined with Machine Learning

Lazebnik [6] is the most directly relevant precedent: an ABM integrating epidemiological and economic models, generating synthetic outcomes used to train an ML model evaluated with SHAP analysis to predict near-optimal firm strategy. This validates, in a peer-reviewed context, the core architectural decision that ABM-generated synthetic data is a defensible substitute for firm-level ground truth when the latter does not exist.

D. SME-Specific and Decision-Framework Studies

Two 2026 studies extend this literature toward the SME context specifically, supporting the premise that SME resilience decisions cannot be modeled as a smaller version of large-firm decisions. Bruzzone et al. [7] surveyed the state of the art in ABM for supply chain risk management; a 2024 systematic review confirmed ABM remains the dominant paradigm for this problem class, with ML-augmented ABM an emerging extension.

E. Synthesis and Research Gap

No reviewed study combines explicit capital and storage constraints as the strategy-determining mechanism, a controlled diversification-versus-buffering comparison, varying disruption severity and duration, and grounding in a verified Philippine food-manufacturing context. This combination is the contribution of this study.

III. METHODOLOGY

A. Research Design

This is a scenario-based, mechanism-testing agent-based model, not a digital twin or firm-level empirical model. It tests a theoretical mechanism situated in a Philippine food-manufacturing context to make the temporal and firm-size environment realistic.

B. Data Sources

Tier 1 (empirical): demand calibration uses PSA MISSI VaPI for food manufacturing, January 2020 through June 2026 (78 months), spanning two base-year eras (2000=100 through Dec 2020, 2018=100 from Jan 2021), detrended separately via STL [8] and combined into multiplicative seasonal factors. March is the strongest peak (approx. 1.147); month-over-month volatility is approximately 7.89%. Firm-size and capital calibration uses the DTI/PSA MSME classification: Small (P3,000,001-P15,000,000 assets), Medium (P15,000,001-P100,000,000), sampled log-uniformly across the combined range.

Tier 2 (literature-derived and scenario assumptions): parameters unavailable from Philippine public data are intended to derive from peer-reviewed literature or documented scenario assumptions. However, the twelve

ranges in Table I are not individually traceable to a specific cited source in the current draft; no per-parameter citation ties, for example, supplier reliability (0.60-0.98) to a named study. This is a documented gap between the stated sourcing method and what is actually shown, and every Tier 2 range should be read as a scenario assumption rather than literature-verified until a parameter-by-parameter source table is added.

C. Model Design

The ABM represents an SME firm agent, heterogeneous supplier agents, and an aggregate demand process calibrated to MISSI seasonality. The firm agent carries capital, storage capacity, inventory, and production capacity; each supplier carries a reliability parameter and stochastic delivery function.

Two strategies are implemented as configurable behaviors on matched scenarios. Diversification: the firm establishes a backup supplier in month 1, paying onboarding cost upfront; the backup only transacts during disruption, at a cost premium, with exposure independent of the primary suppliers' disruption. Buffering: the firm builds and maintains stock, bounded by storage capacity, during non-disrupted months, and draws it down during disruption without seeking alternate suppliers.

Capital is decremented by every cost incurred and floored at zero; once exhausted, no further spending occurs, but existing inventory is preserved. Discretionary purchases include a partial-purchase fallback when capital is insufficient but still positive.

D. Scenario Matrix and Sampling

The parameter space spans twelve dimensions (Table I), sampled via Latin Hypercube Sampling. Wide-range parameters are sampled log-uniformly given the study's focus on capital-constrained regimes.

TABLE I
Scenario Parameter Space

Parameter	Range	Scale
Capital	P3M to P100M	Log
Storage capacity	50-1,500 units	Linear
Production capacity	50-200 units/mo	Linear
Supplier count	1 to 4	Linear
Supplier reliability	0.60-0.98	Linear
Disruption severity	0.10-0.95	Linear
Disruption duration	0.33-12 months	Log
Diversification onboarding	P100-P50,000	Log
Diversification premium	1%-50%	Linear
Buffering holding cost	0.2%-10%/mo	Log
Unit cost	P20-P500	Linear
Opex (frac. of capital)	0.25%-2.5%/mo	Linear

The full sweep generates 1,000 scenarios crossing all twelve dimensions, giving the downstream ML stage a continuous, well-distributed feature space.

E. Simulation Procedure

For each scenario, the model runs under both strategies with Monte Carlo replication (100 repetitions in the full design) to average out single-run stochastic variance. Both strategies share the same random seed per scenario, ensuring identical demand realizations pre-disruption; random sequences diverge once disruption begins, since diversification's backup mechanism consumes additional draws. This is a partial, not complete, paired-comparison design.

F. Outcome Metrics

- Service level: proportion of monthly demand fulfilled from available inventory.
- Recovery time: first month at which service level reaches 95%+ for three consecutive months post-disruption.
- Total cost and ending capital: cumulative spending and capital remaining at 24 months.
- Preferred strategy: the higher unrounded mean service level across repetitions, with a tolerance band below which scenarios are classified equal.

G. Machine Learning Layer

A Random Forest is trained on the twelve scenario parameters to predict preferred strategy and service-level-gap magnitude. SHAP analysis [9] identifies which parameters most strongly drive predicted outcomes, addressing RQ5. This layer characterizes the simulation's own behavior; it is not a claim about real-world SME outcomes, consistent with the precedent in [6].

IV. MODEL PIPELINE

The full pipeline proceeds as follows: (1) load the PSA MISSI VaPI series and compute STL-based seasonal factors and volatility; (2) sample 1,000 scenarios via Latin Hypercube Sampling across the twelve-parameter space; (3) construct a shared parameter set per scenario used identically by both strategies; (4) run the ABM for 24 months under each strategy, replicated 100 times via Monte Carlo sampling; (5) aggregate replications into mean outcome metrics per scenario; (6) compare aggregated outcomes and assign a preferred-strategy label; (7) compile all scenarios into a single synthetic dataset; (8) train a Random Forest and apply SHAP analysis to identify dominant drivers. Each stage's output is the next stage's input; no stage other than the model itself introduces new assumptions once initial parameter ranges are set.

V. MODEL VALIDATION AND SIMULATION UPDATES

This section documents the independent audit this model underwent prior to analysis, presented as a methodological strength: reported findings survived scrutiny rather than being accepted at first pass.

A. Issues Identified and Resolved

TABLE II

Issue	Resolution
Unbounded negative capital	Floored at zero; inventory preserved
PSA data integrity (2021+)	Re-extracted cell by cell, spot-checked
Date range scope drift	Trimmed to Jan 2020 - Jun 2026
Unrealistic capital range	Replaced with DTI/PSA MSME range
Backup supplier coupling	Decoupled from primary severity
Unpaired strategy comparison	Shared seed per scenario
Seasonal decomposition method	Replaced moving average with STL

B. Effect on Headline Results

Table III tracks how the strategy-preference breakdown changed as each issue was resolved.

TABLE III

Model version	Div.	Buf.	Equal
Original	224	357	419
Capital floor fix only	69	292	639
+Data fix, old range	8	156	836
Current (final)	117	402	481

The largest driver of change was the capital-range correction: under the unrealistic P1-P500,000 range, 85.2% of scenarios exhausted capital before disruption occurred, structurally disabling diversification. Under the DTI/PSA-calibrated range, 0% hit the capital floor, and diversification's win rate rose substantially, indicating much of the originally reported disadvantage was a parameter artifact rather than a finding about the strategy itself.

C. Remaining Open Items

- Disruption severity and duration remain swept rather than fixed to a single historical event, preserving the ability to answer RQ3-RQ4.
- The 12-month duration upper bound does not fully bracket the most severe disruption episode in the calibration window.
- The paired-seed comparison is partial; full pairing would require separating demand and supplier-delivery random streams.
- The model excludes a revenue mechanism; capital is monotonically depleting across the horizon.
- The twelve Tier 2 parameter ranges are not currently traceable to individual cited sources despite being described as literature-derived.

VI. RESULTS (PRELIMINARY DEMONSTRATION RUN)

Results in this section come from a scaled-down demonstration run: 100 scenarios at 20 Monte Carlo repetitions per strategy, rather than the full 1,000-by-100 design. This section demonstrates the mechanism behaves sensibly and gives preliminary, directional answers to RQ1-RQ5; subgroup sizes are small enough that patterns should be read as suggestive, not statistically conclusive.

A. Overall Outcome

Across 100 scenarios, buffering was preferred in 85, diversification in 12, and 3 were statistically equal (0.5-point tolerance). Mean service level was 91.3% under buffering versus 82.8% under diversification; average ending capital was nearly identical (approx. P19.95M diversification vs. P19.83M buffering), indicating buffering's advantage was not achieved by spending less.

B. RQ1: Capital Constraint Tiers

Diversification's win count rose only slightly across capital tiers (2 Low, 4 Medium, 6 High of approx. 33-34 each) -- a weak, gradual trend, consistent with capital's low Random Forest importance (1.5%).

C. RQ2: Storage Capacity Constraints

Storage showed a clearer pattern: diversification won 8 of 34 in the Low tier versus 1 and 3 in Medium and High. This is consistent with diversification becoming comparatively more valuable when a firm cannot physically store enough to buffer through a disruption.

D. RQ3: Disruption Severity

Severity tiers showed no monotonic pattern (3, 7, and 2 diversification wins across Low, Medium, High), consistent with severity's minor Random Forest importance (0.7%); the full sweep would be needed to determine whether this reflects the true mechanism or limited sample size.

E. RQ4: Disruption Duration

Duration produced the clearest result, aligning with its dominant Random Forest importance (60.3%).

TABLE IV

Tier	Range (mo)	Buf. wins	Div. wins	Equal
Short	1	34	0	0
Medium	1-4 (avg 2.1)	33	0	0
Long	4-12 (avg 7.1)	18	12	3

Diversification never won at 4 months or less; all 12 wins occurred in the Long tier (avg. 7.1 months), consistent with buffering's fixed stockpile eventually exhausting under prolonged disruption while diversification's backup supplier continues delivering.

F. RQ5: Crossover Region and Sensitivity

The Random Forest achieved a held-out R-squared of 0.916. Disruption duration (60.3%) and supplier reliability (31.0%) together account for over 91% of feature importance; all ten remaining parameters combined account for under 9%. This uses Random Forest feature importance as a substitute for the originally specified SHAP analysis, which was not installable in the offline build environment, and should be treated as preliminary.

VII. DISCUSSION AND CONCLUSION

A. Interpretation

Within this demonstration's scope, duration -- not severity, capital, or storage alone -- most strongly determines which strategy performs better. Buffering suffices for short and moderate disruptions; diversification's advantage emerges once a disruption outlasts a buffering firm's stockpile. Storage capacity showed a secondary, weaker version of the same logic.

B. Relevance to SDG 9

These preliminary results offer a testable planning heuristic: firms facing short, low-severity interruptions may be better served by inventory buffering, while firms in industries prone to prolonged disruption, or with limited storage, may find diversification more resilient despite higher cost -- directly supporting more informed SME resilience investment decisions.

C. Limitations of This Demonstration Run

- Scale: 100 scenarios at 20 repetitions, not the full 1,000-by-100 design.
- Parameter sourcing: 11 of 12 ranges are not individually traceable to cited sources.
- supplier_count is sampled but not read anywhere in the simulation logic.
- The preferred-strategy label uses service level alone; ending capital and recovery time are computed but unused in that decision.
- STL and SHAP were substituted with simpler equivalents due to offline package availability.
- Calibration used Total Manufacturing VaPI/VoPI rather than the Food Manufacturing division.

D. Recommended Next Steps

- Run the full 1,000-by-100 sweep to confirm these directional patterns at the intended statistical power.
- Build a parameter-by-parameter source table for the eleven unsourced Tier 2 parameters.
- Wire supplier_count into the simulation or remove it from the swept space.
- Extend the preferred-strategy criterion to jointly consider capital and recovery time.
- Install statsmodels and shap in a networked environment to restore the originally specified methods.

E. Conclusion

This demonstration shows the capital-and-storage-constrained resilience mechanism produces coherent, interpretable results consistent with its own internal logic: buffering favored under short disruptions and ample storage, diversification favored under long disruptions and constrained storage. These results are preliminary and scoped to a reduced run with documented substitutions and sourcing gaps. They support proceeding to the full 1,000-scenario study and should not be cited as final findings until that run, a proper SHAP analysis, and a completed source table are in place.

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